OS Security

The CIA Triad (Core Security Principles)

no		???
1	**Confidentiality**	Only authorized people can access data. Example: Password protection Encryption
2	**Integrity**	Data cannot be altered or tampered with by unauthorized users. Example: File hashes Digital signatures
3	**Availability**	Systems and data must be accessible when needed. Example: Servers staying online Protection from crashes or DDoS attacks

real example

Confidentiality	Bank accounts require login + encryption so others cannot see your balance.
Integrity	Git commits / digital signatures ensure code hasn't been altered
Availability	Cloud backups and load balancers keep websites online during traffic spikes

Common Threats to CIA

C : Data Breach, PW leaks
I : File Tampering, malware
A : DDos Attack

memory tricks

Confidentiality → Keep secrets

Integrity → Keep data correct

Availability → Keep systems running

Operating systems are critical because they:

- control hardware
- run applications
- protect sensitive data
- If the OS is insecure, everything on the device becomes vulnerable.